PROJECT QUARTZ

Intellectual Property Due Diligence Report - QX-Clock Patent Family

Prepared by Whitfield Carter LLP for Arkwright Quantum Limited

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1. Executive summary

This report summarises our review of the QX-Clock patent family and the associated know-how proposed to be licensed to Caldera Defence Systems Limited under Project Quartz. We have reviewed the patent register extracts, prosecution files supplied by Hartington IP, the inventor assignment chain and the relevant research and collaboration agreements. Subject to the points in section 4, the family is owned by Arkwright Quantum Limited free of third-party security, and is capable of being licensed on an exclusive basis within the defence and aerospace field, subject to one pre-existing non-exclusive research licence in favour of Meridian Diagnostics GmbH which is carved out of the grant.

2. The patent family

The QX-Clock family comprises five patents and applications, summarised below. All are recorded in the name of Arkwright Quantum Limited following the 2022 assignment of the founders rights.

Ref	Jurisdiction	Title (abbrev.)	Status
QX-1	GB	Chip-scale optically-pumped atomic clock	Granted (GB2598417B)
QX-2	EP	Microfabricated vapour cell with anti-relaxation coating	Granted (EP3742210B1)
QX-3	US	Quantum RF magnetometer for navigation	Granted (US11,418,209 B2)
QX-4	JP	Temperature-compensated atomic frequency reference	Pending (examination)
QX-5	EP	Integrated photonic interrogation of vapour-cell clocks	Pending (published)

3. Title and encumbrances

The inventor assignments from the three founder-inventors (Dr Eleanor Frost, Dr Raj Menon and Dr Caroline Webb) to the Company were executed on 4 April 2022 and are valid. We have reviewed the debenture granted to Clydesford Bank plc and confirm that it does not contain a fixed charge over the registered intellectual property; the QX-Clock family is therefore unencumbered for the purposes of the licence. No co-ownership or government march-in rights apply.

4. Issues and recommendations

- Meridian research licence. A non-exclusive, field-limited research licence granted to Meridian Diagnostics GmbH in 2021 covers QX-2 for non-defence metrology use only. It must be expressly carved out of the exclusive grant to Caldera. Recommended: include as a Permitted Encumbrance in the licence and reflect in the warranties.
- QX-4 (JP) examination. The Japanese application is under examination with an outstanding office action due 28 February 2026. Recommend a specific disclosure and a warranty qualification.
- Export control. The technology is rated under UK strategic export controls (PL5001 dual-use). The licence should contain an export-control compliance clause and condition first deliveries on the necessary licences.
- Know-how. The transferred know-how (vapour-cell fabrication recipes and interrogation firmware) is documented in the Technology Transfer Schedule; ensure it is fully scheduled and that departing-employee confidentiality is intact.

5. Conclusion

The QX-Clock family is suitable to support the proposed exclusive licence on the terms contemplated by the term sheet, subject to the carve-outs and warranty qualifications above being reflected in the long-form licence and disclosure letter.